

Dhinesh Sadhu Subramaniam Ponnarasan

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EDUCATION

State University of New York at Binghamton | Clark Fellowship

Binghamton, NY

Master of Science - Information Systems with Applied Data Science; *Cumulative GPA: 3.21/4.00*

Aug 2024 – Present

• Coursework: Machine Learning, Generative AI, Python, JavaScript, API Development, Natural Language Processing, Deep Learning.

Vellore Institute of Technology

Bangalore, India

Post Graduate Program - Data Science; *Cumulative GPA: 3.36/4.00*

Aug 2023 – June 2024

• Coursework: Data Structures and Algorithms, Big Data, Artificial Intelligence, C/C++ Programming, Database, Data visualization

Sri Krishna Arts and Science College

Coimbatore, India

Bachelor of Computer Applications; *Cumulative GPA: 3.20/4.00*

Apr 2019 – Jun 2022

EXPERIENCE

AI/ML & Applications Development Intern

Aug 2025 – Present

Uplify AI

Austin, Texas, United States

- Designed and conducted **scalable research projects** using **Python, R**, and **statistical methodologies** to analyze learning patterns across 500K+ user interactions, applying **quantitative field research** techniques including **regression**, structural equation modeling, and **data mining** to uncover perception drivers that improved engagement by 42%.
- Performed **thorough data analysis** leveraging **SQL databases, R**, and **Python** to manage and analyze large datasets, identifying patterns and translating **research into actionable insights** that enhanced operational performance, collaborating with **cross-functional teams** including engineers, product managers, and business stakeholders.
- Applied **AI/ML tools (TensorFlow, NLP, machine learning)** and **scientific research principles** to develop recommendation models achieving 88% accuracy, conducting **experiments** and iteratively improving methods based on **quantitative metrics** and feedback to deliver measurable customer impact.
- Communicated **data-driven insights and recommendations** through **written documents** and presentations to both technical and non-technical stakeholders, demonstrating excellent **written communication** skills while engaging in discussions to ensure research generates **actionable findings** for strategic decision making.

Software Development Engineer

June 2022 – July 2023

V3Techserv

Chennai, India

- Conducted **end-to-end research** investigating feasibility of applying **scientific principles and concepts to business problems**, utilizing **R, Python**, and **statistical methodologies** including **forecasting techniques** and **regression analysis** that improved prediction accuracy by 34% through systematic experimentation.
- Leveraged **SQL databases** and **data analysis packages (R, Python)** to manage and analyze 2M+ records, performing **data mining**, exploratory analysis, and **advanced modeling** to understand relationships between key metrics, diving deep into large amounts of data to uncover actionable patterns.
- Translated **research, data, and advanced analytics into strategic direction**, creating comprehensive documentation and presentations that communicated complex findings to diverse stakeholders, demonstrating creative problem-solving and ability to tailor research that informs decision making and strategic tactics.

Software Developer

Nov 2021 – May 2022

Freelancer

Coimbatore, India

- Applied **scientific research and analysis** methodologies using **Python, R**, and **SQL** to design experiments analyzing 100K+ transaction records, implementing **data mining techniques** and statistical modeling that increased conversion by 28% through evidence-based insights and pattern identification.
- Utilized **statistical methodologies** and **data analysis packages (Python, R, SQL)** to understand relationships between customer behavior metrics, conducting thorough analysis and creating visualizations that translated research into **actionable recommendations** for business optimization.
- Collaborated with **cross-functional teams** in fast-paced environment to define requirements, establish success metrics, and deliver solutions, demonstrating self-starter mentality, comfort with ambiguity, and ability to work semi-autonomously on complex problems with visible risks.

PROJECTS

Scalable Learning Analytics & Predictive Modeling Research Platform

Jan 2024 – Mar 2024

Python, R, SQL, Statistical Modeling, Regression, Data Mining, ML, Quantitative Research

- Designed and conducted **scalable research project** using **Python, R**, and **SQL databases** to analyze 1M+ learning interactions, applying **statistical methodologies** including **regression, forecasting**, and structural equation modeling to understand perception drivers, achieving 89% predictive accuracy through systematic experimentation.
- Performed **thorough data analysis** leveraging **data mining techniques** and **advanced modeling skills** to identify patterns in learner behavior, diving deep into large datasets and translating **research into actionable insights** that enhanced operational performance by 94%, iteratively improving methods based on **quantitative metrics**.
- Communicated **data-driven insights and recommendations** through comprehensive **written documents** and presentations to technical and non-technical stakeholders, demonstrating excellent **written communication** and creative problem-solving to inform strategic decision making and drive measurable customer impact.

Educational Technology Research: NLP & ML for Learning Insights

May 2023 – Jul 2023

Python, R, SQL, NLP, Machine Learning, Statistical Analysis, Data Mining, Experimentation

- Conducted **end-to-end research** investigating feasibility of applying **scientific principles** to learning technology, utilizing **NLP, machine learning**, and **AI/ML tools** with **Python** and **R**, achieving 85% F1-score through systematic **experimentation** and evaluation of emerging methodologies.
- Applied **statistical methodologies** and **data analysis packages (R, Python, SQL)** to manage and analyze 500K+ text records, using **data mining** and **forecasting techniques** to understand relationships between content features and learning outcomes, reducing analysis time by 60% through efficient processing workflows.
- Collaborated with **cross-functional teams** to define requirements and establish success metrics, translating **research and advanced analytics into strategic direction**, creating actionable recommendations through clear written communication that informed stakeholder decision making in fast-paced, ever-changing environment.

TECHNICAL SKILLS

Scientific Research & Analysis: Quantitative Field Research, End-to-End Research, Scientific Research Design, Experimentation, Scalable Research Projects, Research Methodologies, Scientific Principles Application

Data Analysis Tools: R, Python, SAS, Matlab, SQL, SQL Databases, Data Analysis Packages, Large Dataset Management, Data Mining Techniques

Statistical Methodologies: Regression, Structural Equation Modeling, Forecasting, Statistical Modeling, Hypothesis Testing, Statistical Analysis, Probability, Advanced Modeling Skills

Machine Learning & AI: AI/ML Tools, TensorFlow, Scikit-learn, Machine Learning Processing, NLP (Natural Language Processing), NLU, Computer Vision, PyTorch, Model Evaluation

Data Science & Analytics: Data Mining, Advanced Analytics, Pattern Identification, Perception Drivers Analysis, Predictive Modeling, Quantitative Metrics, Performance Analysis

Research Translation: Actionable Insights, Data-Driven Recommendations, Strategic Direction, Research into Action, Stakeholder Decision Making, Business Problem Solving

Communication Skills: Written Communication, Written Documents, Presentations, Technical & Non-Technical Communication, Documentation, Data Visualization, Storytelling with Data

Collaboration & Leadership: Cross-Functional Teams, Stakeholder Engagement, Engineers Partnership, Product Collaboration, Business Teams, Self-Starter, Semi-Autonomous Work

Data Processing: NumPy, Pandas, SciPy, ETL Pipelines, Data Wrangling, Data Cleaning, Large-Scale Data Processing, Apache Spark

Tools & Platforms: Git, Jupyter, Tableau, Power BI, Excel, PostgreSQL, Data Visualization, Operations Research, Optimization

PUBLICATIONS

Unsupervised Anomaly Detection for Network Intrusion Using Deep Autoencoders (ICNGCS 2025, IEEE)

- Conducted **quantitative research** applying **scientific principles** and **machine learning** to security challenges, achieving 91% accuracy through systematic experimentation, demonstrating end-to-end research skills and data-driven insights communication.

Collaboration Search with Knowledge Sharing and Summarization (ICSES 2024, IEEE)

- Designed **scalable research project** leveraging **NLP** and **machine learning processing**, translating research into actionable system improvements, collaborating with cross-functional teams to deliver measurable impact.

Image to Audio Conversion to Aid Visually Impaired People by CNN (ICESC 2023, IEEE)

- Applied **scientific research methodologies** and **AI/ML tools** to accessibility challenges, conducting thorough analysis and experiments, communicating findings through written documentation for technical and non-technical audiences.